

SYLLABUS :-

Name of the subject Geomatics(New Number) 2. L-T-P 3-0-0 3. Credits 3 4. Name of the Department Civil Engineering 5. Status of the subject (a) Specify the Session, Semester: Autumn Semester (b) Please specify the level of the subject: 2nd year B.Tech. (c) Whether the subject will be offered as core or elective: Core (d) The semester in which the subject will be offered: Autumn (e) Name of the program in whose curricula this subject will be included: B.Tech., Dual Degree in Civil Engg. and B. Arch. 6. Prerequisites for the subject, if any None 7. Objectives The objective is to make students familiar with the basic engineering spatial measurements needed to plan and construct an infrastructure. The name of the course has been modified to 'Geomatics' from 'Surveying.' Geomatics includes the tools and techniques used in land surveying, remote sensing, cartography, geographic information systems (GIS), global-navigation satellite systems (GPS, GLONASS, Galileo, Compass), photogrammetry, geophysics, geography and related forms of earth mapping. As such 'Surveying' is a subset of 'Geomatics.' 8. Contents (42 hours) Topic Hours Basic concepts of Geomatics: A brief intro. to the concept of surveying, mapping, remote sensing and GIS; Necessity of Geomatics; Basic measurements; Units of measurement. 3 Basic Principles and Types of surveying and introduction to the common tools of surveying: Traversing; Triangulation, Trilateration, closing errors in traversing and their adjustment. 3 Distance Measurement: Chain and Tapes; Optical Distance Measurement; Electronic Distance Measurement; Errors and corrections in Distance Measurement, considerations for earth curvature and refraction. 6 Angular Measurement: Direction, Azimuth, Meridian; Compass; Sources of Errors, Propagation of errors. 3 Plane Table Surveying, types; 2-point problem and 3- point problem. 3 - 126 - Levelling and Contouring: principles of simple and differential levelling; Introduction to dumpy level and electronic level instruments; Determination of heights; Block levelling; Characteristics and methods of contouring; Sources of errors and their corrections. 6 Theodolite survey, Repetition and reiteration methods, latitudes and departures; Tacheometry 6 Introduction to remote sensing and GIS techniques, LIDAR, INSAR. Introduction to GPS, surveying using DGPS and remote sensing techniques, survey using drones, construction survey. 9 Total Station: Operating principle, link with linear, angular and vertical measurements. Text books 1. Surveying and Levelling by NN Basak, McGraw Hill Education, 2nd edition 2. Surveying Vol. 1 and Vol. 2 by SK Duggal, McGraw-Hill, Fifth edition Reference books 1. Surveying and Levelling Vol. 1 and Vol. 2 by TP Kanetkar and SV Kulkarni, Pune Vidyarthi Griha Prakashan 2. Surveying Vol. 1 and Vol. 2 by BC Punmia, Laxmi Publications; Sixteenth edition